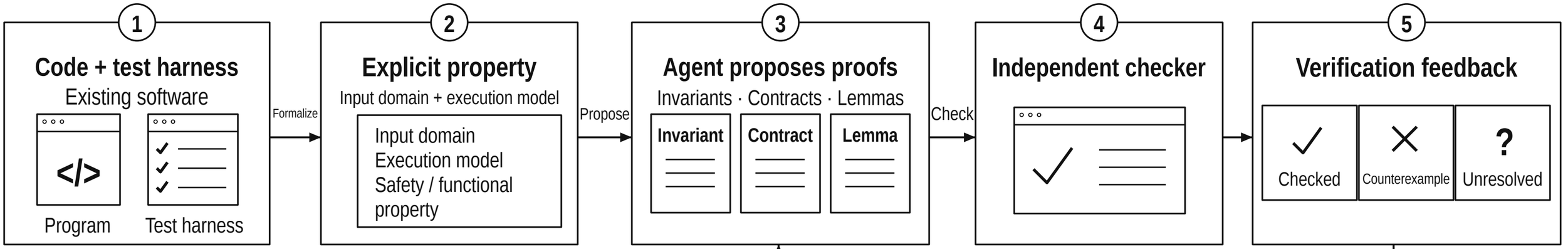
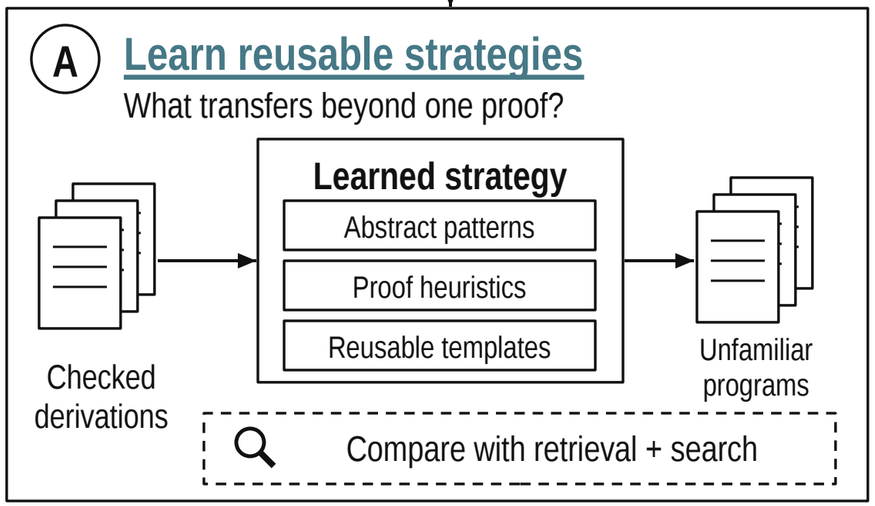
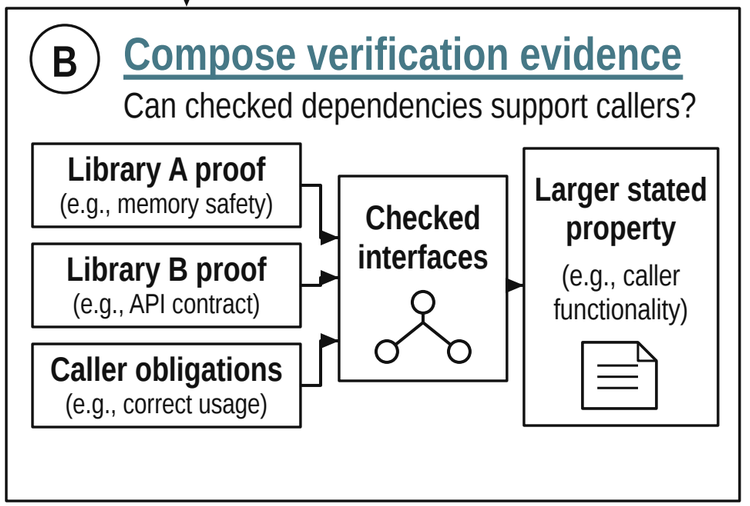
Learning Reusable Formal Reasoning

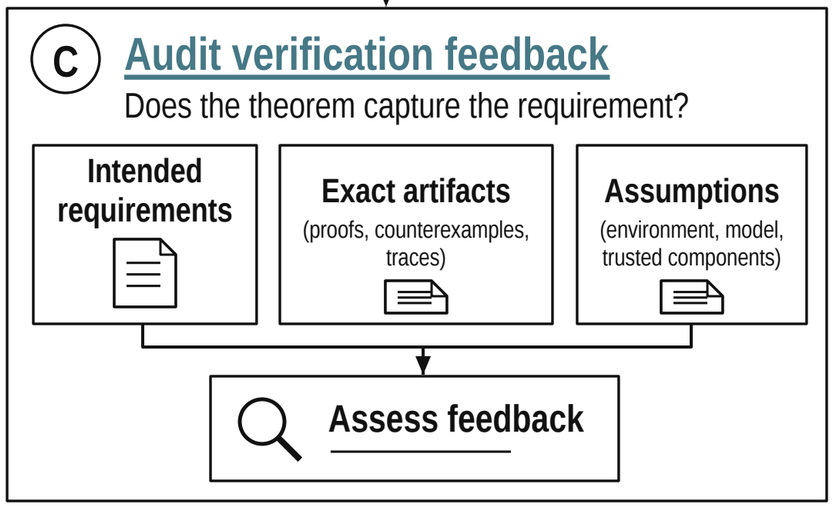
Proposed research

From expensive proofs to transferable reasoning









Efficient, reusable, independently checkable verification

Guarantees depend on the property, input domain, execution semantics, and trusted checking.

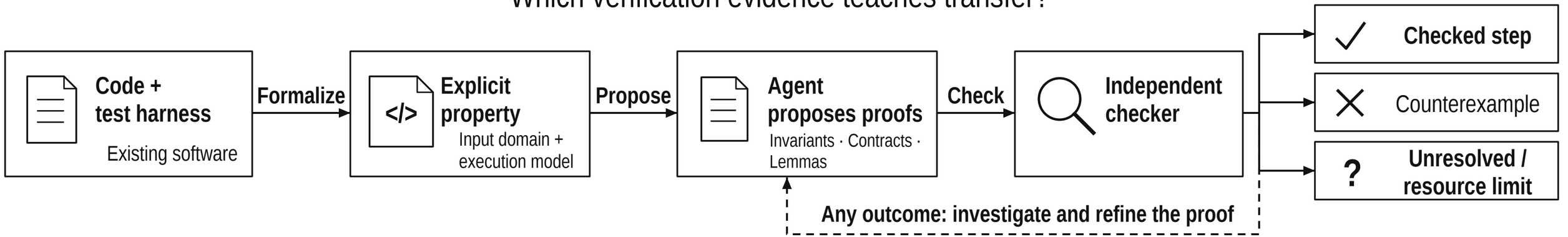
[1] Anthropic + Theorem (2026)

[2] Theorem (2026)

Learning Reusable Proof Strategies

Proposed research

Which verification evidence teaches transfer?



Compare three experimental mechanisms

1 Retrieve

Reuse checked proof artifacts

Search and reuse from existing invariants, lemmas, etc.

2 Refine

Adapt during one task

Iteratively improve using checker feedback on the same task.

3 Learn

Train reusable proof strategies

Strategy

Compare final proofs, derivations, and revisions.

Held-out transfer test

Unfamiliar programs + changed obligations

Matched data, tools, and total cost

Sharing a proof and learning a strategy are evaluated separately.

Checked success does not establish generalization.

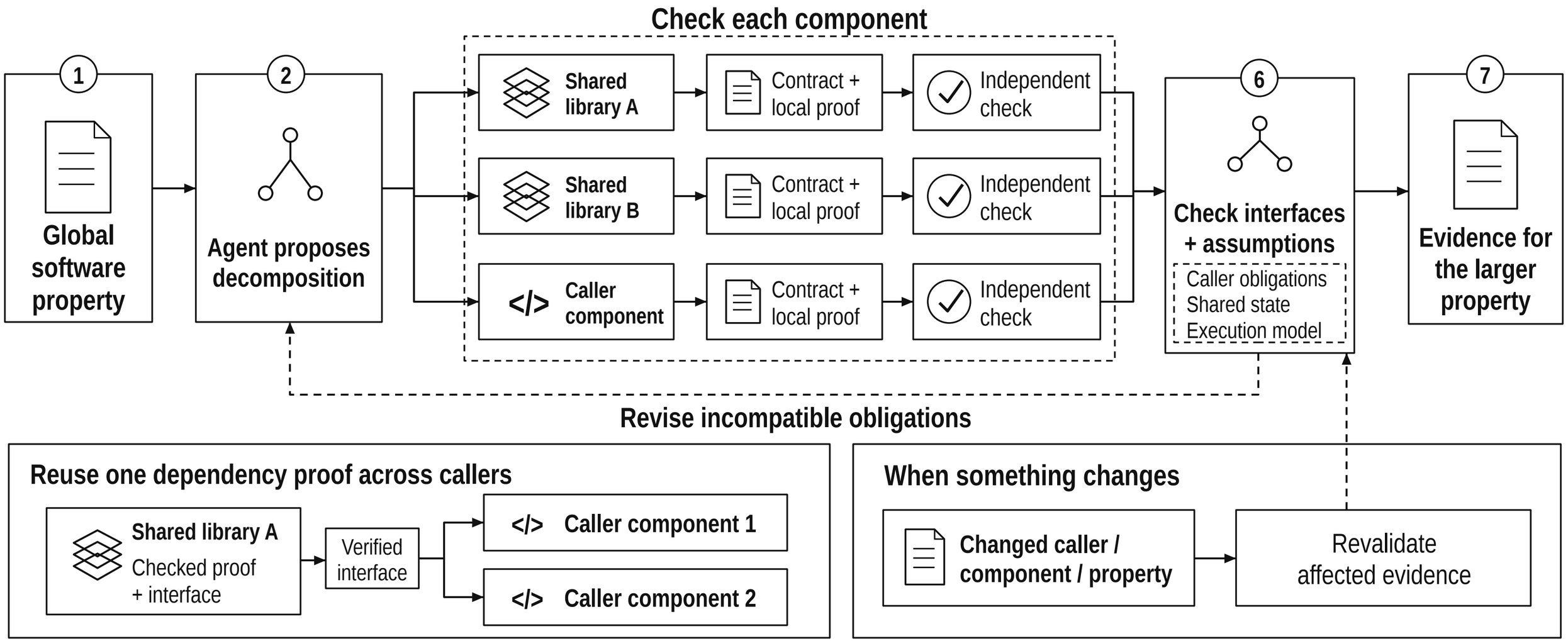
[1] Anthropic + Theorem (2026)

[2] Theorem (2026)

Compositional Verification

Proposed research

Reuse checked components through verified interfaces



Reuse is conditional on matching interfaces, artifacts, and assumptions.

Larger guarantees require checked interfaces and explicit assumptions.

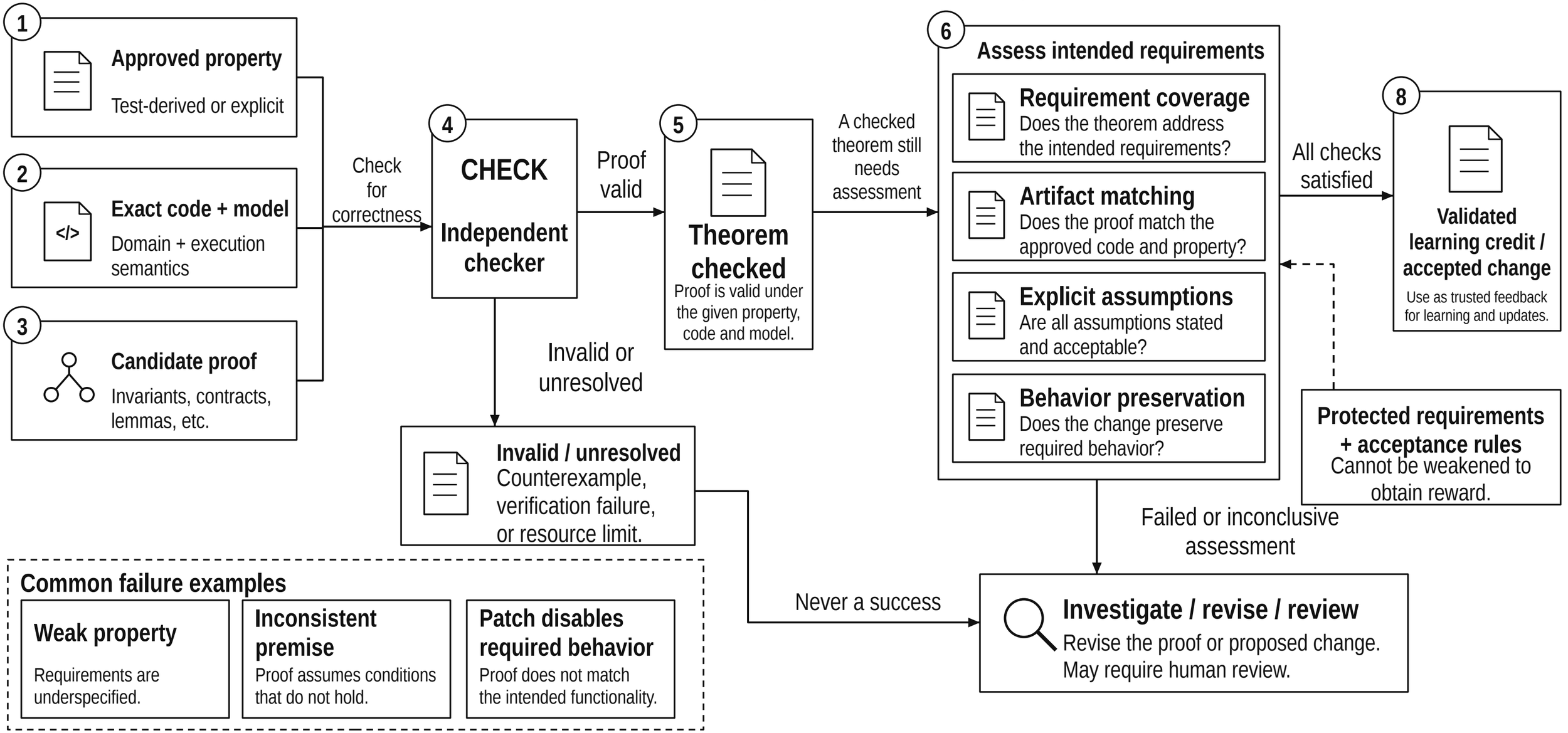
[1] Anthropic + Theorem (2026)

[2] Theorem (2026)

Trustworthy Verification Feedback

Proposed research

A checked theorem must support the intended requirement



A valid proof establishes its theorem; intended requirements need separate assessment.

[1] Anthropic + Theorem (2026)

[2] Theorem (2026)